

# Literature-implied answer: the optimal uniform-POVM reverse Holder constant

Source: Aubrun–Lancien, arXiv:1309.6003.

Supporting theorem: Murawski, arXiv:2211.05210, Theorem 1.3.

**Status.** `literature_implied_answer (full source subquestion)`. This note answers the optimal-constant question following Proposition 4.1 of Aubrun–Lancien. The supporting paper does not explicitly cite that POVM question; the implication is through log-concave moment comparison.

**Source question.** In Section 4.1, after Proposition 4.1 and equations (10)–(12), Aubrun–Lancien prove

$$\frac{1}{\sqrt{18}} \|\Delta\|_{2(1)} \leq \|\Delta\|_{U_d} \leq \|\Delta\|_{2(1)}$$

for every Hermitian  $\Delta \in H(\mathbb{C}^d)$ , where

$$\|\Delta\|_{2(1)} = (\text{Tr}(\Delta^2) + (\text{Tr } \Delta)^2)^{1/2}.$$

They observe that this is a reverse Holder inequality and ask for the optimal constant, noting that  $\sqrt{18}$  is presumably far from optimal.

**Identification.** Let  $g$  be a standard complex Gaussian vector in  $\mathbb{C}^d$ . As observed in the source,

$$\|\Delta\|_{U_d} = \mathbb{E}|\langle g, \Delta g \rangle|, \quad \|\Delta\|_{2(1)} = (\mathbb{E}|\langle g, \Delta g \rangle|^2)^{1/2}.$$

Diagonalizing  $\Delta$  gives

$$\langle g, \Delta g \rangle = \sum_j \lambda_j E_j,$$

where the  $E_j$  are independent exponential random variables of mean one. This law is log-concave, because the product exponential density on the positive orthant is log-concave and log-concavity is preserved under linear images (Prekopa–Leindler).

Murawski’s Theorem 1.3 states that, among all real log-concave random variables, the ratio  $\|X\|_p/\|X\|_q$  for  $p > q > 0$  is maximized by a shifted exponential distribution. Taking  $(p, q) = (2, 1)$  therefore reduces the optimal constant to a one-variable shifted-exponential calculation.

**Theorem.** *Let  $s_0$  be the unique positive solution of*

$$e^s = 2(s^2 - s + 1).$$

*Set*

$$\kappa = \frac{s_0 - 1 + 2e^{-s_0}}{\sqrt{s_0^2 - 2s_0 + 2}} = 0.63500423058334237014 \dots$$

*Then for every  $d$  and every Hermitian  $\Delta \in H(\mathbb{C}^d)$ ,*

$$\kappa \|\Delta\|_{2(1)} \leq \|\Delta\|_{U_d} \leq \|\Delta\|_{2(1)}.$$

*The constant  $\kappa$  is optimal uniformly over all dimensions.*

**Calculation of the shifted-exponential constant.** Let  $E$  have exponential distribution with mean one. For  $s \geq 0$ ,

$$\mathbb{E}|E - s| = s - 1 + 2e^{-s}, \quad (\mathbb{E}|E - s|^2)^{1/2} = \sqrt{s^2 - 2s + 2}.$$

For  $s < 0$  the ratio  $\|E - s\|_2/\|E - s\|_1$  is smaller than the maximum obtained below. Hence the extremal lower constant is the minimum over  $s \geq 0$  of

$$r(s) = \frac{s - 1 + 2e^{-s}}{\sqrt{s^2 - 2s + 2}}.$$

A direct derivative computation gives the critical equation

$$e^s = 2(s^2 - s + 1).$$

The function  $e^s - 2(s^2 - s + 1)$  has positive derivative on  $[0, \infty)$  after the elementary check that its derivative has positive minimum, so there is a unique positive solution  $s_0$ , and it gives the global minimum since  $r(0) = 1/\sqrt{2}$  and  $r(s) \rightarrow 1$  as  $s \rightarrow \infty$ .

**Sharpness inside the POVM matrix class.** For  $N \geq 1$ , take a Hermitian matrix with eigenvalues

$$1, -s_0/N, \dots, -s_0/N$$

in dimension  $N + 1$ . Then

$$\langle g, \Delta_N g \rangle = E_0 - \frac{s_0}{N} \sum_{j=1}^N E_j.$$

By the law of large numbers, this converges in  $L^1$  and  $L^2$  to  $E_0 - s_0$ . Therefore no constant larger than  $\kappa$  can hold uniformly for all Hermitian matrices and all dimensions.

**Scope limitations.** This answers the optimal-constant subquestion in Section 4.1 of Aubrun–Lancien. It does not address the separate questions about deterministic sparsification, removing the logarithmic factor in arbitrary zonoid approximation, or upgrading arbitrary-POVM sub-POVM sparsification to a true POVM.

## References

- [1] G. Aubrun and C. Lancien, *Zonoids and Sparsification of Quantum Measurements*, arXiv:1309.6003.
- [2] D. Murawski, *Comparing moments of real log-concave random variables*, arXiv:2211.05210.